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Question	Answer	Marks
1	**Defining the problem**	
	θ is the independent variable and x is the dependent variable **or** vary θ and measure x	**1**
	keep (angle) β constant	**1**
	Methods of data collection	
	labelled diagram of workable experiment including: • spring attached at both ends e.g. one end connected to a clamp and stand • strip free to move • at least two labels from: clamp, stand, wire, strip, spring, bench (Do not accept extra masses added to strip.)	**1**
	use a rule to measure L and d	**1**
	use a protractor to measure θ **or** use a rule to measure appropriate distances to determine θ by trigonometry methods	**1**
	measure original length of spring and new length of spring using rule/calipers	**1**
	Method of analysis	
	plot a graph of x against cos θ **or** cos θ against x (Allow log x against log (cos θ).)	**1**
	relationship is valid if a straight line passing through the origin is produced (Allow straight line with gradient = 1 for log-log graph.)	**1**
	for x against cos θ: $W = \frac{\text{gradient} \times 2kd \sin\beta}{L}$ **or** for cos θ against x: $W = \frac{2kd \sin\beta}{\text{gradient} \times L}$	**1**

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Question	Answer	Marks
1	**Additional detail including safety considerations**	**6**
	D1 wear goggles <u>to prevent spring/wire/strip entering into eyes</u> **or** (retort) stand used to support spring is clamped to bench	
	D2 keep distance *d* <u>constant</u>	
	<u>D3 description</u> of (separate) experiment to determine *k*, e.g. weigh mass and measure extension	
	D4 *k* = weight / extension **or** *mg* / extension **or** gradient of weight–extension graph for candidate's workable (separate) experiment	
	D5 method to prevent strip at point P sliding, e.g. use adhesive putty/hinge (Do not accept methods that prevent rotation at point P.)	
	D6 use fiducial markers on spring at both ends **or** measure length of spring on both sides and average	
	D7 method to attach wire to strip, e.g. wire wrapped around the strip/(strong) tape/drill hole and tie wire	
	D8 determine *x* by subtracting original length of spring from new length	
	D9 adjust support of spring to keep β constant	
	D10 protractor correctly positioned on diagram to measure θ **or** correct trigonometric relationship given for θ	